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PROJECT PROPOSAL

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SupportBrain AI

A RAG-Enabled Multi-Agent SaaS Platform
for Intelligent Omnichannel Customer Support

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Abstract

Customer support is one of the most critical yet resource-intensive operations for modern businesses. Organizations are expected to provide continuous assistance across websites, social media platforms, messaging applications, and voice communication while maintaining fast response times and high customer satisfaction. Traditional support centers require large teams of human agents, resulting in significant operational costs and inconsistent service quality.

This project proposes the development of **SupportBrain AI**, a cloud-based Software-as-a-Service (SaaS) platform that leverages Artificial Intelligence, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), AI Agents, and Voice Intelligence to automate customer support. Businesses can simply provide their official website URL, documentation, PDF manuals, or social media pages, allowing the system to automatically extract relevant knowledge, generate vector embeddings, and build an intelligent knowledge base capable of answering customer queries accurately.

The platform supports omnichannel communication including website live chat, Messenger, WhatsApp, voice conversations, and image-based queries. It also incorporates human handoff mechanisms for complex situations, enabling seamless transition from AI agents to human representatives. Business owners will have access to dashboards for conversation monitoring, analytics, subscription management, team administration, and secure API key management.

The proposed solution aims to reduce operational costs, improve customer satisfaction, provide uninterrupted 24/7 support, and demonstrate the practical integration of modern AI technologies within enterprise customer service systems.

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1. Introduction

1.1 Background

In today’s digital economy, customer support has become one of the most important services offered by businesses. Customers expect immediate, accurate, and personalized responses regardless of the time or communication platform they use. Modern organizations interact with customers through websites, mobile applications, social media platforms, messaging applications, emails, and voice calls. Providing consistent support across these channels requires a significant number of human agents, making customer service one of the most resource-intensive departments within an organization.

With the rapid advancement of Artificial Intelligence (AI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and autonomous AI Agents, businesses now have the opportunity to automate repetitive customer interactions while maintaining high-quality responses. Unlike traditional chatbots that rely on predefined rules or frequently asked questions (FAQs), AI-powered support systems can understand natural language, retrieve relevant knowledge, and generate context-aware responses.

Despite these technological advancements, many small and medium-sized businesses still face difficulties implementing intelligent support systems due to high development costs, technical complexity, and the need for continuous maintenance. Existing solutions often require manual knowledge base creation, extensive AI training, or dedicated technical teams.

This project proposes **SupportBrain AI**, a cloud-based Software-as-a-Service (SaaS) platform designed to simplify AI-powered customer support. Business owners only need to provide their company website, product documentation, PDF manuals, or social media pages. The platform automatically extracts business information, constructs a searchable knowledge base using Retrieval-Augmented Generation (RAG), and enables AI agents to answer customer questions through multiple communication channels including websites, Messenger, WhatsApp, and Voice AI.

The proposed system aims to reduce operational costs, improve customer satisfaction, provide uninterrupted 24/7 customer support, and make advanced AI technologies accessible to businesses of all sizes.

1.2 Problem Statement

Customer support remains one of the biggest operational challenges for modern businesses. Companies must hire and train multiple support representatives to answer repetitive customer questions throughout the day. Maintaining a large support team significantly increases operational expenses while still failing to provide continuous 24-hour service.

Traditional chatbot systems are often limited to predefined rules and cannot understand complex customer questions or retrieve information from multiple knowledge sources. Furthermore, businesses frequently update their products, pricing, services, and policies, requiring continuous manual updates to chatbot knowledge bases.

Another challenge is the increasing number of communication channels. Customers expect support through websites, Facebook Messenger, WhatsApp, emails, and voice calls. Managing all these channels separately leads to inconsistent customer experiences and increased administrative workload.

Many AI solutions available today require organizations to manually prepare datasets, train AI models, or possess advanced technical expertise. These requirements make AI adoption expensive and inaccessible for many organizations, particularly small and medium-sized enterprises.

Therefore, there is a need for an intelligent, scalable, and user-friendly SaaS platform capable of automatically creating a business knowledge base from publicly available resources such as websites, documentation, and social media pages while providing intelligent customer support across multiple communication channels.

1.3 Project Motivation

Recent developments in Large Language Models and Retrieval-Augmented Generation have demonstrated that AI systems can significantly improve customer support quality while reducing operational costs. However, many existing platforms are either expensive, difficult to configure, or limited to a single communication channel.

The motivation behind this project is to develop an affordable and scalable AI-powered customer support platform that businesses can configure within minutes. Instead of manually training AI models, business owners should simply provide their website URL or business documents, allowing the system to automatically generate an intelligent knowledge base.

Additionally, integrating voice communication, image understanding, messaging applications, and human handoff within a single platform provides businesses with a unified customer support ecosystem. Such an integrated solution can improve customer satisfaction, reduce response times, and increase business productivity.

1.4 Project Objectives

The primary objective of this project is to design and develop an AI-powered customer support platform capable of automating business communication using Retrieval-Augmented Generation, AI Agents, and Voice Intelligence.

The specific objectives are as follows:

1. Develop a cloud-based multi-tenant SaaS customer support platform.
2. Automatically collect business knowledge from websites, PDF documents, and social media pages.
3. Generate vector embeddings for intelligent semantic information retrieval.
4. Implement a Retrieval-Augmented Generation (RAG) pipeline for accurate AI responses.
5. Integrate AI chat agents for website customer support.
6. Provide Voice AI support using speech recognition and text-to-speech technologies.
7. Support communication through Messenger, WhatsApp, and other messaging platforms.
8. Implement Human Handoff for complex customer queries.
9. Design an administration dashboard for managing organizations, support teams, and AI knowledge bases.
10. Provide secure API key management and subscription-based SaaS functionality.
11. Evaluate the performance of the proposed system based on response accuracy, response time, scalability, and user satisfaction.

1.5 Project Scope

The scope of the proposed project covers the design, development, deployment, and evaluation of an intelligent AI-powered customer support platform.

The system will include the following major functionalities:

- Business registration and authentication.
- Multi-organization (multi-tenant) support.
- Automatic website content extraction.

- Document upload and processing.
- Knowledge Base creation using vector embeddings.
- Retrieval-Augmented Generation (RAG).
- AI-powered website chatbot.
- Voice-based customer support.
- Messenger and WhatsApp integration.
- Human handoff to live support agents.
- Conversation history management.
- Analytics dashboard.
- Team and permission management.
- Subscription and billing management.
- Secure API key storage.
- Error monitoring and logging.

The first version of the project will primarily focus on customer support automation for web-based businesses. Advanced enterprise integrations such as ERP systems, CRM synchronization, custom AI model training, and video call support are considered future enhancements and are outside the scope of this project.

1.6 Expected Outcomes

Upon successful completion of this project, the proposed system is expected to achieve the following outcomes:

- A fully functional AI-powered SaaS customer support platform.
- Intelligent customer support using Retrieval-Augmented Generation.
- Automatic business knowledge generation from websites and documents.
- Unified customer communication across multiple platforms.
- Improved customer response quality.
- Reduced customer support operational costs.

- Continuous 24/7 AI-assisted customer service.
- Secure and scalable cloud-based architecture.
- Production-ready deployment suitable for real-world business environments.

1.7 Project Questions

This project aims to investigate the following research questions:

1. Can Retrieval-Augmented Generation improve the accuracy of AI-powered customer support compared to traditional rule-based chatbots?
2. How effectively can AI agents automate customer support using business websites as the primary knowledge source?
3. Does integrating multiple communication channels into a unified SaaS platform improve customer experience?
4. Can intelligent human handoff reduce customer dissatisfaction when AI confidence is low?
5. Is the proposed platform scalable enough to support multiple organizations simultaneously?

2. Proposed System

2.1 Overview

SupportBrain AI is a cloud-based Software-as-a-Service (SaaS) platform that enables businesses to automate customer support using Artificial Intelligence, Retrieval-Augmented Generation (RAG), AI Agents, and Voice Intelligence. The platform is designed to reduce operational costs while providing fast, accurate, and uninterrupted customer service.

Unlike conventional chatbot systems that rely on manually written responses or pre-defined rules, the proposed system automatically builds an intelligent knowledge base by extracting information from business websites, PDF documents, product manuals, and social media pages. Once the knowledge base is created, AI agents retrieve relevant information and generate context-aware responses using Large Language Models (LLMs).

The platform supports multiple communication channels including website live chat, Facebook Messenger, WhatsApp, voice conversations, and image-based queries. It also provides an administrative dashboard where business owners can manage knowledge sources, monitor conversations, configure AI models, and manage support staff.

2.2 System Workflow

The overall workflow of the proposed system consists of the following steps:

1. Business owner creates an organization account.
2. The business owner submits a website URL, Facebook page, or uploads business documents.
3. The system automatically crawls and extracts business information.
4. Extracted data is cleaned and converted into vector embeddings.
5. The embeddings are stored inside a Vector Database to create an intelligent knowledge base.

6. When a customer asks a question, the AI Agent retrieves the most relevant information using Retrieval-Augmented Generation (RAG).
7. The selected Large Language Model generates an accurate response.
8. If the AI confidence level is below a predefined threshold, the conversation is transferred to a human support representative.
9. The administrator can monitor conversations and system performance through the dashboard.

2.3 Major Features

The proposed system provides the following major features.

2.3.1 AI Customer Support

The platform provides intelligent conversational support capable of understanding natural language questions and generating context-aware responses using Large Language Models.

2.3.2 Automatic Website Knowledge Extraction

Businesses only need to provide their website URL. The system automatically crawls pages, extracts useful information, and prepares it for AI processing.

2.3.3 Document Processing

Business owners can upload PDF files, manuals, FAQs, and other documents. These documents are processed automatically and added to the knowledge base.

2.3.4 Knowledge Base using RAG

The extracted information is transformed into vector embeddings and stored in a Vector Database. Retrieval-Augmented Generation ensures that AI responses are based on business-specific information rather than general knowledge.

2.3.5 Voice AI Support

Customers can communicate using voice messages. Speech is converted into text, processed by the AI Agent, and the response is converted back into speech.

2.3.6 Image Understanding

Customers can upload screenshots or product images. The AI analyzes the image and combines visual understanding with the knowledge base to generate accurate responses.

2.3.7 Omnichannel Communication

The proposed platform supports multiple communication channels including:

- Website Live Chat
- Facebook Messenger
- WhatsApp
- Voice Communication
- Email (Future Enhancement)

2.3.8 Human Handoff

If AI cannot confidently answer a customer query, the conversation is automatically transferred to an available human support representative.

2.3.9 Conversation Analytics

Business owners can monitor:

- Total Conversations
- AI Resolution Rate
- Average Response Time
- Customer Satisfaction
- Human Handoff Rate

2.3.10 Subscription Billing

The system supports SaaS subscription plans with organization-level usage management.

2.4 High-Level System Components

The proposed platform consists of several independent but interconnected modules.

2.4.1 Authentication Module

Responsible for:

- User Registration
- Login
- Role-Based Access Control
- Organization Management

2.4.2 Knowledge Base Module

Responsible for:

- Website Crawling
- PDF Processing
- Facebook Content Extraction
- Embedding Generation
- Vector Database Storage

2.4.3 AI Agent Module

Responsible for:

- Prompt Management
- Context Retrieval
- Tool Calling
- AI Reasoning
- Conversation Memory

2.4.4 RAG Engine

Responsible for:

- Semantic Search
- Similarity Matching
- Knowledge Retrieval
- Context Injection
- Response Generation

2.4.5 Communication Module

Responsible for:

- Website Chat Widget
- Messenger Integration
- WhatsApp Integration
- Voice Support
- Future API Integrations

2.4.6 Support Dashboard

Provides administrators with:

- Live Conversations
- Human Agent Assignment
- Conversation History
- Ticket Management
- Customer Details

2.4.7 Analytics Module

Provides:

- Customer Statistics
- AI Performance
- Response Accuracy
- Conversation Reports
- Usage Monitoring

2.4.8 Billing Module

Responsible for:

- Subscription Plans
- Usage Tracking
- Organization Limits
- Payment History

2.5 User Roles

The proposed system supports multiple user roles.

2.5.1 Super Administrator

Responsible for:

- Platform Management
- Organization Approval
- Subscription Management
- AI Configuration

2.5.2 Business Owner

Responsible for:

- Creating Organizations
- Managing Team Members
- Uploading Knowledge Sources
- Monitoring Analytics
- Managing AI Settings

2.5.3 Support Agent

Responsible for:

- Receiving Human Handoff Requests
- Solving Customer Problems
- Closing Conversations

2.5.4 Customer

Customers can:

- Ask Questions
- Upload Images
- Send Voice Messages
- Receive AI Responses
- Request Human Support

2.6 Functional Requirements

The proposed system shall:

1. Allow organizations to register and authenticate.
2. Automatically crawl business websites.
3. Extract useful business information.
4. Generate embeddings for semantic search.
5. Store embeddings in a Vector Database.
6. Answer customer questions using Retrieval-Augmented Generation.
7. Support text, image, and voice-based communication.
8. Integrate with Messenger and WhatsApp.
9. Transfer conversations to human agents when necessary.
10. Maintain conversation history.
11. Provide dashboards for administrators.
12. Generate business analytics.
13. Manage subscriptions and organization plans.

3. System Architecture

3.1 Overview

The proposed system follows a modern cloud-native multi-tier Software-as-a-Service (SaaS) architecture. The architecture is designed to be modular, scalable, secure, and fault tolerant. Each module performs an independent responsibility while communicating through secure APIs. This architecture enables easy maintenance, future expansion, and support for multiple organizations (multi-tenancy).

The system consists of four major layers:

1. Presentation Layer
2. Application Layer
3. Artificial Intelligence Layer
4. Data Layer

The interaction between these layers allows customer requests to be processed efficiently while maintaining high performance and reliability.

3.2 Overall System Architecture

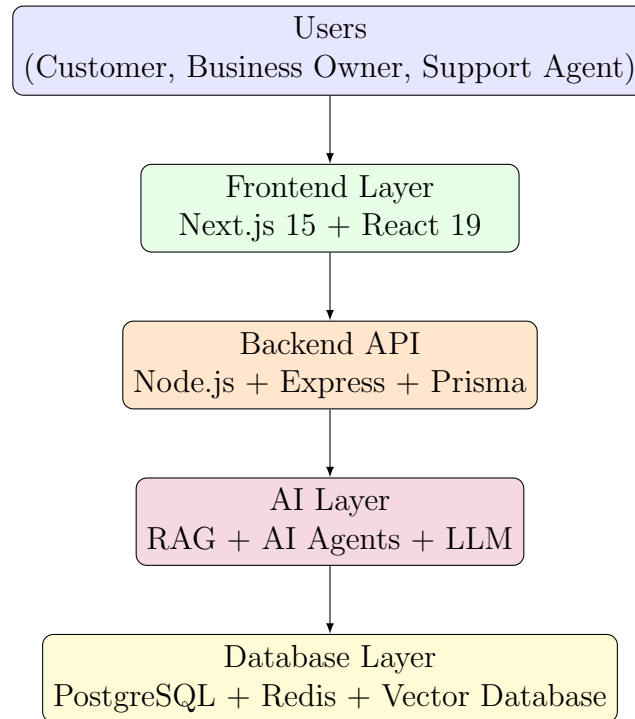


Figure 3.1: High-Level System Architecture

3.3 System Components

The proposed platform consists of the following interconnected components.

3.3.1 Presentation Layer

The Presentation Layer provides user interaction through a responsive web application developed using Next.js 15 and React 19. Different dashboards are available for administrators, business owners, support agents, and customers.

Major responsibilities include:

- User Interface
- Authentication
- Chat Widget
- Dashboard
- Analytics
- Team Management

3.3.2 Application Layer

The Application Layer contains the core business logic of the platform.

Its responsibilities include:

- Authentication
- Organization Management
- Conversation Management
- Knowledge Base Management
- Billing
- API Integration
- Notification Services

3.3.3 Artificial Intelligence Layer

The AI Layer is responsible for intelligent customer support.

Its primary components are:

- Large Language Models
- AI Agents
- Retrieval-Augmented Generation
- Embedding Generator
- Prompt Engineering
- Conversation Memory
- Tool Calling

3.3.4 Data Layer

The Data Layer stores all persistent information.

Major storage technologies include:

- PostgreSQL
- Redis
- Vector Database
- AWS S3

3.4 Artificial Intelligence Workflow

The AI workflow begins when a customer submits a question through any supported communication channel. The request is forwarded to the AI Agent, which retrieves the most relevant information from the organization's knowledge base before generating the final response.

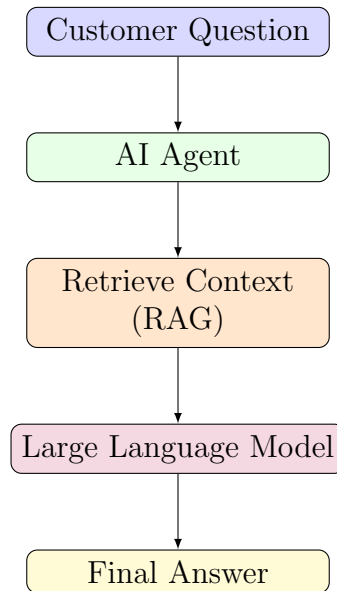


Figure 3.2: Artificial Intelligence Workflow

3.5 Retrieval-Augmented Generation (RAG) Pipeline

Retrieval-Augmented Generation is the core intelligence mechanism of the proposed system. Instead of relying solely on the general knowledge of a Large Language Model, the AI retrieves organization-specific information from a Vector Database before generating a response.

The workflow consists of the following steps:

1. Customer submits a question.
2. The question is converted into embeddings.
3. Similarity search is performed.
4. Relevant documents are retrieved.
5. Retrieved context is combined with the original question.
6. Large Language Model generates the final response.
7. Response is delivered to the customer.

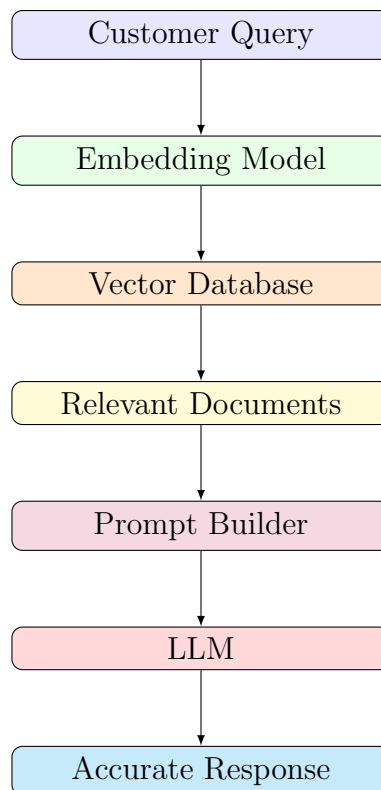


Figure 3.3: Retrieval-Augmented Generation Pipeline

3.6 Component Diagram

The proposed platform contains several independent services that communicate through REST APIs.

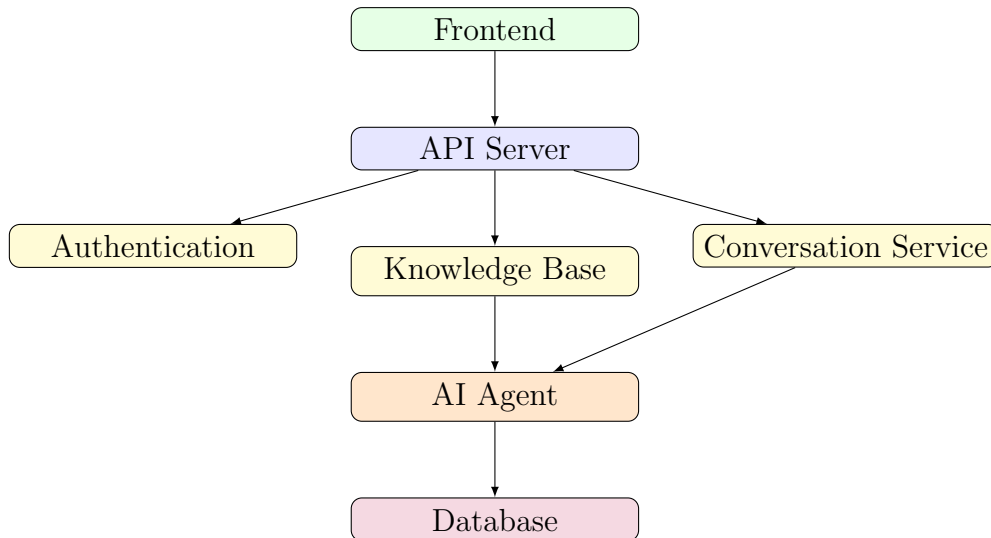


Figure 3.4: Major Software Components

3.7 Data Flow Description

The complete processing sequence of the proposed platform is summarized below:

1. The business owner registers an organization.
2. Business knowledge is collected from websites, documents, or social media pages.
3. The crawler extracts useful information.
4. The embedding service converts textual information into vectors.
5. The vectors are stored in the Vector Database.
6. Customers submit questions through chat, voice, or messaging applications.
7. The AI Agent retrieves relevant information using the RAG pipeline.
8. The selected Large Language Model generates the final response.
9. If the AI confidence score is low, the request is automatically transferred to a human support representative.
10. All conversations are stored for future analytics and continuous system improvement.

4. Technology Stack

4.1 Overview

The proposed system will be developed using modern web technologies, cloud services, and Artificial Intelligence frameworks. The selected technologies are widely adopted in industry due to their scalability, maintainability, security, and performance.

4.2 Frontend Technologies

Technology	Version	Purpose
Next.js	Latest	Full-stack React Framework
React	Latest	User Interface Development
Tailwind CSS	Latest	Responsive Styling
shadcn/ui	Latest	Modern UI Components
TypeScript	Latest	Type Safety

Table 4.1: Frontend Technologies

4.3 Backend Technologies

Technology	Purpose
Node.js	Runtime Environment
Express.js	REST API Development
Python	Web Crawl/Scrapping
Prisma ORM	Database ORM
PostgreSQL	Relational Database
Redis	Caching and Queue
AWS S3	File Storage
AWS Secrets Manager	Secure API Key Storage

Table 4.2: Backend Technologies

4.4 Artificial Intelligence Technologies

Technology	Purpose
OpenAI GPT	Large Language Model
Anthropic Claude	Alternative AI Model
Grok	Multi-Model Support
LangChain	AI Orchestration
LangGraph	AI Agent Workflow
Pinecone/Qdrant	Vector Database
OpenAI Embeddings	Semantic Search
RAG	Knowledge Retrieval

Table 4.3: AI Technologies

4.5 Development Tools

- Visual Studio Code
- GitHub
- Docker
- Turborepo
- Postman
- Sentry
- Vercel
- AWS

4.6 Technology Justification

The selected technology stack enables rapid development while ensuring scalability, modularity, and maintainability. Next.js and React provide a modern frontend experience, Node.js supports high-performance backend services, PostgreSQL ensures reliable data storage, while Retrieval-Augmented Generation combined with AI Agents enables accurate business-specific responses.

5. Development Methodology

5.1 Project Development Roadmap

Since the proposed system is a large-scale AI-powered SaaS platform and is being developed by a single developer as a final-year project, the implementation has been divided into three academic semesters. Each semester focuses on a well-defined milestone to ensure gradual development, continuous evaluation, and successful project completion.

Table 5.1: Three-Semester Development Roadmap

Semester	Project Phase	Major Activities	Deliverables
Summer 2026	Planning, Design & Prototype	Literature Review Requirement Analysis Software Requirement Specification UI/UX Design Database Design System Architecture Frontend Development Interactive Prototype	Project Proposal Complete Frontend UI Prototype System Design Database Schema
Fall 2026	Backend Development & AI	Authentication Organization Management REST API Knowledge Base Website Scraper PDF Processing Embedding Generation Vector Database Retrieval-Augmented Generation AI Agent Dashboard	Working SaaS Backend Knowledge Base AI Chat Admin Dashboard Integrated System
Spring 2027	Deployment, Testing & Optimization	Voice AI Messenger Integration WhatsApp Integration Human Handoff Subscription Billing Security Performance Optimization Cloud Deployment Testing Documentation	Production System Deployment Final Thesis Project Demonstration Performance Report

5.2 Major Project Milestones

Table 5.2: Project Milestones

No.	Milestone	Status
1	Requirement Analysis	Planned
2	UI/UX Design	Planned
3	Frontend Development	Planned
4	Backend Development	Planned
5	Knowledge Base Development	Planned
6	AI Agent Integration	Planned
7	RAG Pipeline Implementation	Planned
8	Voice AI Integration	Planned
9	Messenger & WhatsApp Integration	Planned
10	Deployment & Testing	Planned
11	Final Project Submission	Planned

5.3 Development Methodology

The proposed project follows the Agile Software Development methodology. Development activities are divided into small iterative phases where planning, implementation, testing, documentation, and supervisor feedback are continuously incorporated. This methodology reduces development risk while allowing incremental delivery of project features throughout the three academic semesters.

6. Conclusion

SupportBrain AI proposes a modern Artificial Intelligence-based customer support platform capable of transforming traditional support centers into intelligent automated systems. By combining Retrieval-Augmented Generation, AI Agents, Voice Intelligence, and cloud-native SaaS architecture, the proposed solution enables businesses to provide accurate, scalable, and continuous customer service across multiple communication channels. Unlike conventional chatbot systems, the proposed platform automatically generates a knowledge base from business websites, documents, and social media content, significantly reducing manual configuration efforts. Human handoff mechanisms ensure seamless collaboration between AI and human support representatives, improving both customer satisfaction and operational efficiency. The project is planned over three academic semesters to ensure systematic development, implementation, testing, deployment, and evaluation. This phased approach allows the development of a production-ready software system while maintaining academic quality and engineering standards. Future enhancements may include multilingual support, CRM integration, ERP connectivity, predictive analytics, AI-powered workflow automation, sentiment analysis, and support for additional communication platforms. The successful completion of this project will demonstrate the practical application of Artificial Intelligence in enterprise customer support while contributing to modern intelligent business automation systems.